CS 148a PS3

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24 February 2026

§1 Convolutional Neural Networks

§1.1 Why skip connections prevent vanishing gradients

(i) $\frac{\partial \mathcal{L}}{\partial w^{(1)}}$

By the chain rule, we expand the gradient of the loss with respect to $w^{(1)}$:

$$\frac{\partial \mathcal{L}}{\partial w^{(1)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial h^{(L)}}{\partial h^{(L-1)}} \cdot \frac{\partial h^{(L-1)}}{\partial h^{(L-2)}} \cdots \frac{\partial h^{(2)}}{\partial h^{(1)}} \cdot \frac{\partial h^{(1)}}{\partial w^{(1)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \left(\prod_{l=2}^{L} \frac{\partial h^{(l)}}{\partial h^{(l-1)}} \right) \frac{\partial h^{(1)}}{\partial w^{(1)}}$$

(ii) $\frac{\partial h^{(l)}}{\partial h^{(l-1)}}$

Since $h^{(l)} = \sigma\big(w^{(l)} h^{(l-1)} + b^{(l)}\big)$, by the chain rule:

$$\frac{\partial h^{(l)}}{\partial h^{(l-1)}} = \sigma'\big(w^{(l)} h^{(l-1)} + b^{(l)}\big)\, w^{(l)}$$

Substituting: With $h^{(l-1)} = 0.1$, $w^{(l)} = 0.1$, $b^{(l)} = 0$:

$$z^{(l)} = w^{(l)} h^{(l-1)} + b^{(l)} = 0.1 \cdot 0.1 + 0 = 0.01$$

Since $\sigma = \text{ReLU}$ and $z^{(l)} = 0.01 > 0$, $\sigma'(z^{(l)}) = 1$. Therefore:

$$\frac{\partial h^{(l)}}{\partial h^{(l-1)}} = 1 \cdot 0.1 = \boxed{0.1}$$

(iii) Effect on $\frac{\partial \mathcal{L}}{\partial w^{(1)}}$ for large L

The gradient contains the product $\prod_{l=2}^{L} \frac{\partial h^{(l)}}{\partial h^{(l-1)}}$.

- If $\left| \frac{\partial h^{(l)}}{\partial h^{(l-1)}} \right| < 1$:
  - for every layer (i.e. 0.1 from above), the product behaves as $(0.1)^{L-1}$, which decays exponentially toward 0 as L grows. This is the vanishing gradient problem. The gradient signal for early layers becomes very small so the weights receive almost no update and training.
- If $\left| \frac{\partial h^{(l)}}{\partial h^{(l-1)}} \right| > 1$:
  - for every layer, the product behaves as $\forall X > 0, (1+X)^{L-1}$, which grows exponentially as L increases. This is the exploding gradient problem. The gradient signal for early layers becomes extremely large so the weights receive excessively large updates, leading to instability/divergence during training.

(iv) Skip connections alleviate vanishing gradients

With skip connections, $h^{(l)} = \sigma\big(w^{(l)} h^{(l-1)} + b^{(l)}\big) + h^{(l-1)}$. Differentiating:

$$\frac{\partial h^{(l)}}{\partial h^{(l-1)}} = \sigma'\big(w^{(l)} h^{(l-1)} + b^{(l)}\big)\, w^{(l)} + 1$$

Now we have the additive $+1$ term. Even when $\sigma'(\cdot)\, w^{(l)}$ is very small (i.e. 0.1), each factor is now $1 + 0.1 = 1.1$ rather than 0.1. The product over L layers becomes:

$$\prod_{l=2}^{L} \Big(\sigma'\big(w^{(l)} h^{(l-1)} + b^{(l)}\big)\, w^{(l)} + 1\Big)$$

Because every factor is close to (or greater than) 1, this product no longer vanishes exponentially. In the worst case where the nonlinear contribution is zero (i.e. ReLU in the inactive region gives $\sigma' = 0$), each factor is still exactly 1, so the gradient passes through unattenuated. This is why skip connections effectively allow the loss signal to propagate all the way back to early layers, hence mitigating vanishing gradient.

§2 RNNs and LSTMs

§2.1 (b) Backpropagation through time equations

(i)

$$\boldsymbol{x}_t \in \mathbb{R}^{B\times D}, \quad \hat{\boldsymbol{y}}_t \in \mathbb{R}^{B\times D}, \quad \boldsymbol{h}_t \in \mathbb{R}^{B\times H}, \quad \boldsymbol{W}_{xh} \in \mathbb{R}^{D\times H}, \quad \boldsymbol{W}_{hh} \in \mathbb{R}^{H\times H}, \quad \boldsymbol{W}_{hy} \in \mathbb{R}^{H\times D}$$
$$\boldsymbol{b}_h \in \mathbb{R}^{1\times H}, \quad \boldsymbol{b}_y \in \mathbb{R}^{1\times D}$$

(ii)

Since $\hat{\boldsymbol{y}}_t = \boldsymbol{W}_{hy}\boldsymbol{h}_t + \boldsymbol{b}_y$ and $\mathcal{L} = \sum_t \mathcal{L}_t$:

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{W}_{hy}} = \sum_{t=1}^{T} \boldsymbol{h}_t^\top \frac{\partial \mathcal{L}_t}{\partial \hat{\boldsymbol{y}}_t}, \qquad \frac{\partial \mathcal{L}}{\partial \boldsymbol{b}_y} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t}{\partial \hat{\boldsymbol{y}}_t}$$

(iii)

$$\frac{\partial f}{\partial \beta} = \sum_{i=1}^{n} \frac{\partial f}{\partial \alpha_i} \frac{\partial \alpha_i}{\partial \beta}$$

(iv)

Since $\boldsymbol{h}_t = f(\boldsymbol{h}_1, \boldsymbol{h}_2, \ldots, \boldsymbol{h}_{t-1}, \boldsymbol{W}_{hh})$, we can resucrively apply the multivariate chain rule. We want to note that $\boldsymbol{h}_t$ depends on $\boldsymbol{W}_{hh}$ both directly and through all previous hidden states:

$$\frac{d\boldsymbol{h}_t}{d\boldsymbol{W}_{hh}} = \sum_{k=1}^{t} \left(\prod_{j=k+1}^{t} \frac{\partial \boldsymbol{h}_j}{\partial \boldsymbol{h}_{j-1}} \right) \frac{\partial \boldsymbol{h}_k}{\partial \boldsymbol{W}_{hh}}$$

Note: $\frac{\partial \boldsymbol{h}_k}{\partial \boldsymbol{W}_{hh}}$ represents the direct partial derivative of $\boldsymbol{h}_k$ w.r.t $\boldsymbol{W}_{hh}$. When $k = t$ the product is empty and equals the identity.

(v)

Using $\hat{\boldsymbol{y}}_t = \boldsymbol{W}_{hy}\boldsymbol{h}_t + \boldsymbol{b}_y$ and substituting part (iv):

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{W}_{hh}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t}{\partial \hat{\boldsymbol{y}}_t} \frac{\partial \hat{\boldsymbol{y}}_t}{\partial \boldsymbol{h}_t} \sum_{k=1}^{t} \left(\prod_{j=k+1}^{t} \frac{\partial \boldsymbol{h}_j}{\partial \boldsymbol{h}_{j-1}} \right) \frac{\partial \boldsymbol{h}_k}{\partial \boldsymbol{W}_{hh}}$$

(vi)

Similarly, replacing the direct partial with respect to $\boldsymbol{W}_{xh}$):

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{W}_{xh}} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t}{\partial \hat{\boldsymbol{y}}_t} \frac{\partial \hat{\boldsymbol{y}}_t}{\partial \boldsymbol{h}_t} \sum_{k=1}^{t} \left(\prod_{j=k+1}^{t} \frac{\partial \boldsymbol{h}_j}{\partial \boldsymbol{h}_{j-1}} \right) \frac{\partial \boldsymbol{h}_k}{\partial \boldsymbol{W}_{xh}}$$

(vii)

The term $\prod_{j=k+1}^{t} \frac{\partial \boldsymbol{h}_j}{\partial \boldsymbol{h}_{j-1}}$ leads to the vanishing/exploding gradient problem as described in 1.1.

§2.2 Train LSTM on IMDB Movie Reviews to classify sentiment

(i) The product $I_t \odot \tilde{C}_t$ represents the new information selected by the input gate for storage in long term; I_t controls how much of each memory element is admitted. It is added to $F_t \odot C_{t-1}$ such that the cell state update combines selectively retained old memory (via F_t) with selectively admitted new information.

(ii) Multiplying O_t by $\tanh(C_t)$ allows the LSTM to selectively expose parts of the long term state as the short term hidden state effectively allowing it to choose what's important to remember in the long term and what it can forget; tanh bounds C_t to $[-1, 1]$ to keep values bounded, while the output gate O_t filters out components of the memory that are irrelevant.

§3 Transformers

§3.1 Attention

(i)

$$\text{Attention}(x) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V = \text{softmax}\left(\frac{xW_q(xW_k)^\top}{\sqrt{d_k}}\right)xW_v$$

$$x \in \mathbb{R}^{T\times D}, \quad W_q \in \mathbb{R}^{D\times d_k}, \quad W_k \in \mathbb{R}^{D\times d_k}, \quad W_v \in \mathbb{R}^{D\times d_v}$$

The output shape is $\mathbb{R}^{T\times d_v}$, since $QK^\top \in \mathbb{R}^{T\times T}$ and $V \in \mathbb{R}^{T\times d_v}$.

(ii)

With $T = 1$, $QK^\top$ reduces to $q \cdot k = \sum_{j=1}^{d_k} q_j k_j$. Since $Q = xW_q$ and $K = xW_k \sim \mathcal{N}(0, d_k I)$, each q_j and k_j are independent.

Each $q_j k_j$ has mean 0 and variance 1, so by the Central Limit Theorem,

$$q \cdot k \sim \mathcal{N}(0, d_k).$$

Without scaling,

$$\text{Var}(q \cdot k) = d_k,$$

so the variance of the softmax logits grows linearly with d_k. For large d_k, the logits become large in magnitude and softmax gradients approach 0.

Dividing by $\sqrt{d_k}$ normalizes the variance:

$$\frac{q \cdot k}{\sqrt{d_k}} \sim \mathcal{N}(0, 1).$$

(iii)

For each head $i = 1, \ldots, h$, compute:

$$\mathrm{H}_i = \text{softmax}\left(\frac{xW_q^i(xW_k^i)^\top}{\sqrt{d_k}}\right)xW_v^i.$$

$$\text{MultiHead}(x) = \oplus(\mathrm{H}_1, \ldots, \mathrm{H}_h)W_o.$$

Shapes:

$$W_q^i \in \mathbb{R}^{D\times d_k}, \quad W_k^i \in \mathbb{R}^{D\times d_k}, \quad W_v^i \in \mathbb{R}^{D\times d_v}, \quad W_o \in \mathbb{R}^{hd_v\times D}.$$